

Overview of the Adaptive Immune System

B- and T-cell development, receptor generation, selection, tolerance, and adaptive organization

If you remember nothing else

- B cells and T cells arise from lymphoid precursors but complete development in different locations.
- Adaptive receptors are generated through germline DNA rearrangement, creating enormous receptor diversity.
- Development includes checkpoints that remove, modify, suppress, or eliminate problematic self-reactive clones.
- Failures in developmental genes can produce clinically important immunodeficiency syndromes.

Development framework

- Follow a lymphocyte through: origin -> receptor generation -> developmental checkpoint -> selection/tolerance -> mature cell.
- Receptor generation creates diversity before exposure to the specific antigen.
- Selection is required because random receptor generation can also create self-reactive receptors.

Clinical anchors from the Week 6 guide

- BTK/X-linked agammaglobulinemia is used as a B-cell developmental example.
- DiGeorge syndrome is used as a T-cell developmental example.
- AIRE/APS-1 is used to connect central tolerance with autoimmune disease.

Exam frame

- Development problem: ask which lineage and which checkpoint failed.
- Tolerance problem: ask whether self-reactive cells were inadequately removed or controlled.
- Do not equate receptor diversity with antigen exposure; diversity is generated during development.